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Container for receiving, storing and serving food

Abstract

The invention relates to a container for receiving, keeping warm and/or serving food, wherein the container comprises a base body that can be closed by means of a lid. According to one aspect, a brake is provided, which brakes a closing movement of the lid. According to another aspect, a connecting module for connecting the lid to the base body is provided, wherein the connecting module is detachably connected both to the lid and to the base body.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2003/0218000	12/2002	Haber	N/A	N/A
2005/0279225	12/2004	Chang	N/A	N/A
2008/0010778	12/2007	Tse	16/82	E05F 3/14

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
112144997	12/2019	CN	N/A
69608858	12/2000	DE	N/A
69608858	12/2000	DE	N/A

OTHER PUBLICATIONS

DE69608858 English Translation, Accessed on Mar. 4, 2025 (Year: 2001). cited by examiner
International Search Report issued in Appln. No. PCT/EP2022/050641 mailed May 13, 2022 (2 pages). cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is a national phase entry under 35 U.S.C. § 371 of International Application No. PCT/EP2022/050641, filed Jan. 13, 2022, which claims priority from German Patent Application No. 102021200271.4 filed Jan. 13, 2021, all of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

(2) The invention relates to a container for receiving, keeping warm and/or serving food, wherein the container comprises a base body that can be closed by means of a lid. According to one aspect, a brake is provided, which decelerates (brakes) a closing movement of the lid. According to another aspect, a connecting module for connecting the lid to the base body is provided, wherein the connecting module is detachably connected both to the lid and to the base body.

(3) Containers for receiving, storing and serving food are used, for example, at buffets to keep food warm in a water bath. The containers typically comprise a pan-shaped base body (trough-shaped base body), which can be closed by a lid. The containers are known in a wide variety of embodiments, for example including flat or curved lids, which can be opened or closed as needed to store, and possibly showcase, the foodstuffs stored in the container in a fresh and appealing for as long as possible. These containers are also known, amongst others, in the food service industry under the term “chafing dish.”

(4) One problem with these devices can be the handling of the lid. Removable lids, which are only placed on the base body for closing the same and not connected thereto in an articulated manner, require an additional space for setting it down that must be kept clear. Lids that can be pivoted about horizontal axes and are connected to the base body in an articulated manner, in turn, should be pivotable completely to the rear, for example to be able to remove a water bath vessel. In the process, there is a risk that the server may lose his or her controlled grip of the relatively heavy lids and that the lids are flipped toward the back and down, whereby the vessel on the table could be moved, the water for keeping the food warm could slosh, and food could be spilled. When the lid is being closed, the server must reach far to the rear and down to grab the lid. A need therefore exists for refining the closing mechanism of the lid.

(5) At times, it is desirable to be able to completely demount the lid, which is connected to the base body in an articulated manner, from the base body. For example, it may be desirable to clean the lid separately from the base body in dishwasher. Even though some previously known lids can be removed, these contain mechanical components which are not suited, or are only conditionally suited, for use in dishwashers. Other lids can only be removed from the base body using a tool intended for this purpose. It would be advantageous to find a practical solution for a removable lid.

BRIEF SUMMARY OF THE INVENTION

(6) It is the object of the invention to design a container for receiving, keeping warm and/or serving food that is able to at least partially solve the above-described problems.

(7) This object is achieved by a container having the features of the main claim. Further embodiments of the invention are defined by the subject matter of the dependent claims and the description below.

(8) According to a first aspect, a container, in particular for receiving, keeping warm and/or serving food, is provided. The container comprises: a base body; a lid for closing the base body which can be pivoted between a closed position and an open position by means of a rotary joint (revolute joint); and at least one brake for braking (decelerating) a pivoting movement of the lid and/or for holding the lid in a position between the closed position and the open position.

(9) The brake comprises a freewheel and is configured so as to brake (decelerate) a pivoting movement of the lid in the closing direction, while not braking (decelerating) a pivoting movement of the lid in the opening direction.

(10) The freewheel typically defines a blocking direction and a freewheeling direction. The blocking direction preferably corresponds to the closing direction of the lid and/or the freewheeling direction corresponds to the opening direction of the lid. In other words, the freewheel is coupled to a pivoting movement of the lid in such a way that the brake brakes (decelerates) or holds the lid in the closing direction, not, however, in the opening direction.

(11) The brake can in particular be designed to hold the lid in a plurality of positions, preferably in an infinitely adjustable manner (steplessly), between the closed position and the open position. The

open position denotes the position in which the lid is maximally opened and thus has a maximum opening angle. The maximum opening angle can be established by at least one stop. In this case, the pivoting movement of the lid is thus limited by the stop. In the closed position, the lid completely covers the base body. As a result of the brake, it is thus typically achieved that the lid stays in virtually any position between the closed position and the open position. The user thus only has to move the lid into a position in which the food is accessible, with the lid automatically remaining in this position after being released due to the braking force of the brake. This also means that a minor manual force for overcoming the braking force of the brake is required for closing the lid. This required manual force is usually the greatest in an open vertical state of the lid, for example at approximately 90° , and steadily decreases during closing due to the increasing lever arm. Typically, during the intended use of the container, the lid is oriented substantially horizontally in the closed position, that is, perpendicular to the force of gravity.

(12) Optionally, the brake is a mechanical friction brake. In other words, the braking force is generated by friction between a fixed body and a moving body. Such brakes are typically robust and characterized by a comparatively simple design. In one embodiment, the brake comprises at least one brake pad and a brake disk. It may be provided that the brake disk is arranged so as to be static with respect to the base body. The brake pad can move relative to the base body during a movement of the lid. As an alternative, the brake pad is arranged so as to be static (stationary) with respect to the base body. In this case, the brake disk moves relative to the base body during a movement of the lid.

(13) In some embodiments, the rotary joint (revolute joint) comprises a rotatable shaft that is coupled to the freewheel and connected to the lid. The shaft in general rotates synchronously with a pivoting movement of the lid, that is, both elements are rotated by the same angle of rotation when the lid is being pivoted. Optionally, the brake is configured in such a way that the brake pad and the brake disk are static (stationary) relative to one another in the freewheeling direction of the shaft, and a relative movement of the brake pad and of the brake disk in the blocking direction of the shaft is braked (decelerated). Furthermore, a spring element may be provided, wherein the spring element is configured to push the brake pad and the brake disk together in the axial direction.

(14) The shaft can be surrounded by a bearing bushing. The shaft can be arranged or mounted in an interior space defined by the bearing bushing. The bearing bushing can be connected to at least one cover, which closes the interior space defined by the bearing bushing at least in regions. The brake pad can, for example, be non-rotatably connected to the bearing bushing, for example by way of form fit.

(15) The freewheel can be mounted on the shaft, for example. The freewheel is often configured as a one-way roller clutch. Optionally, the freewheel is arranged radially between the bearing bushing and the shaft. In one embodiment, at least one rolling bearing is arranged radially between the bearing bushing and the shaft, for example two rolling bearings or, if needed, even more than two. As an alternative or in addition, it may also be provided that at least one rolling bearing is integrated into the freewheel. At least one sealing element can be provided, which is configured to seal the freewheel and/or the at least one rolling bearing with respect to water and moisture. The brake can moreover be arranged in a housing, wherein the housing can sometimes comprise two partial housings. For example, the brake can be arranged in or at a connecting module, which connects the lid to the base body. In particular the connecting module described further below is a suitable connecting module.

(16) The brake disk can surround the shaft, the freewheel and/or the bearing bushing in the radial direction. Optionally, a sliding bearing is arranged radially between the bearing bushing and the brake disk, which enables a rotation, preferably a low-friction rotation, of the bearing bushing relative to the brake disk.

(17) Overall, a low-noise braking function and a long service life can be achieved with the brake of the above-described kind. Furthermore, infinitely adjustable lid positioning (steppless lid

positioning) is made possible by the described brake. The brake is furthermore characterized by a small number of components and easy assembly.

(18) According to a second aspect, a further container, in particular for receiving, keeping warm and/or serving food, is described. The container comprises: a base body; a lid for closing the base body which can be pivoted between a closed position and an open position; and a connecting module for connecting the lid to the base body.

(19) The connecting module comprises a first connecting element, which is detachably connected to the lid. Furthermore, the connecting module comprises a second connecting element, which is detachably connected to the base body. The connecting module moreover comprises at least one component for influencing a pivoting movement of the lid.

(20) The expression “detachable connection” shall in particular be understood to mean that the connection can be detached non-destructively and without tools. In this way, the connecting module makes modular coupling of the lid to the base body possible. In particular, the connecting module can be demounted both from the base body and from the lid. Due to the modularity of the coupling, the base body, the lid and the connecting module can be individually replaced, repaired and/or cleaned. For example, the base body and the lid can be cleaned separately from the connecting module. This may in particular be useful when the connecting module or the components thereof, such as the aforementioned component for influencing the pivoting movement of the lid, are sensitive to water, moisture and/or cleaning agents.

(21) The aforementioned component can comprise or be a brake for braking (decelerating) a pivoting movement of the lid and/or for holding the lid in a position between the closed position and the open position. The brake can be configured similarly to the brake described with respect to the first aspect. As an alternative or in addition, the aforementioned component can comprise or be a motor element for moving the lid between the closed position and the open position. The connecting module can comprise a housing in which the component is arranged.

(22) Optionally, the first connecting element comprises a first form fit element, which can be connected to a first counter-piece attached to the lid for forming a first form fit. For example, the first form fit element is configured as a protrusion, while the first counter-piece in a complementary manner comprises a first receptacle having a complementary shape.

(23) In one refinement, a first spring-loaded pressure piece (spring plunger) is provided, which engages in a first locking depression when the first form fit is present. The first spring-loaded pressure piece (spring plunger) can help fix the first form fit. The first spring-loaded pressure piece is installed in the first form fit element, for example. Moreover, the first locking depression can be formed in the first counter-piece. The first locking depression is usually formed in an inner wall of the counter-piece. As an alternative, the first spring-loaded pressure piece can also be installed in the counter-piece. In this case, the first locking depression is formed in the first form fit element.

(24) The first connection can preferably be released starting at an opening angle of the lid of at least 45°, without the base body having to be held in place manually. A spring force or holding force of the first spring-loaded pressure piece and/or the geometry of the first form-fit connection can be designed in such a way that the manual force required for releasing the first connection is generally the lowest at an opening angle of the lid between 80° and 100°, preferably between 85° and 95°, and in particular approximately 90°. For example, at a certain opening angle, in particular 90°, of the lid, a longitudinal axis (or longitudinal direction) of the first form fit element and a longitudinal axis (or longitudinal direction) of the first counter-piece extend parallel to the force of gravity, so that the form-locked connection between the lid and the first connecting element in the open position can be released by lifting of the lid.

(25) Optionally, the second connecting element comprises a second form fit element, which can be connected to a second counter-piece attached to the base body for forming a second form fit. Often times, a second spring-loaded pressure piece (spring plunger) is provided, which latches in a second locking depression when the second form fit is present. The second spring-loaded pressure

piece (spring plunger) can help fix the second form fit.

(26) The second spring-loaded pressure piece can be installed in the second form fit element, for example. Moreover, the second locking depression can be formed in the second counter-piece. The second locking depression is usually formed in an inner wall of the second counter-piece.

(27) As an alternative, the second spring-loaded pressure piece can also be installed in the second counter-piece. In this case, the second locking depression is formed in the second form fit element.

(28) In some embodiments, the second form fit element engages in a second receptacle formed in the counter-piece. The second form fit element is configured as a stud, for example, while the second receptacle has a complementary shape. The second locking depression is usually formed in an inner wall of the second receptacle. Furthermore, the second spring-loaded pressure piece can be installed in the second form fit element. A longitudinal axis of the second spring-loaded pressure piece can extend in particular at an angle, preferably perpendicularly, with respect to a longitudinal axis of the second form fit element. The second locking depression can be configured as a through-opening, so that the second spring-loaded pressure piece can be detached from the outside, preferably by pushing it with a finger.

(29) A third spring-loaded pressure piece (spring plunger), which is preloaded when the second spring-loaded pressure piece is latched, can be provided for releasing the second form fit. The third spring-loaded pressure piece (spring plunger) can be configured to push the second form fit element at least partially out of the receptacle when the second pressure piece is released from the second locking depression. A longitudinal axis of the third spring-loaded pressure piece can be oriented parallel to the longitudinal axis of the second form fit element and/or perpendicularly to the longitudinal axis of the second spring-loaded pressure piece.

(30) In some embodiments, the third spring-loaded pressure piece is installed in the second form fit element. The third spring-loaded pressure piece can rest against the inner wall of the receptacle when the second spring-loaded pressure piece is latched. Optionally, a third depression can be formed in the inner wall of the receptacle, in which the third pressure piece engages.

(31) At least one of the above-described spring-loaded pressure pieces can comprise a sleeve and a spring element arranged in the sleeve, as well as a pressure piece that is partially arranged in the sleeve and partially protrudes from the sleeve, such as a ball or a pin. The spring element is designed to exert a spring force on the pressure piece, whereby the pressure piece is preloaded.

(32) Typically, the first connecting element and the second connecting element are connected to one another by way of a center piece, wherein the component for influencing a pivoting movement of the lid is optionally attached to the center piece or installed in the center piece.

(33) Sometimes, the container can comprise two connecting modules of the described kind. It may be provided that the longitudinal directions of the first form fit elements extend parallel to one another and point in the same direction. The first form fits can also be released simultaneously by pulling on the lid. It may additionally be provided that the longitudinal directions of the second form fit elements run parallel to one another, but are oriented in opposite directions (anti-parallel). As a result, the second form fits cannot be released while the lid is still attached to the connecting modules. Instead, the second form fits (that is, the connection of the connecting modules to the base body) can only be released when the lid has been demounted from the connecting module. This can reduce the likelihood that the lid is cleaned together with the connecting modules.

(34) The base body is often times shaped like a trough (pan). The base body in general comprises a bottom surface, a front wall, a rear wall and two side walls, wherein the connecting module is preferably attached to the rear wall of the base body, and in particular to an upper side of the rear wall. The rear wall can extend in the lateral direction from a first end section to a second end section, wherein the end sections are each connected to a side wall. The connecting modules are preferably attached in the end sections of the rear wall. The reason for this outwardly provided connection of the connecting modules is that the flexural rigidity of the side walls and/or of the corner radii is utilized. In this way, bending of the rear wall while the lid is being opened or closed

can be prevented or reduced. As a result, in turn, additional rear wall reinforcement panels can be omitted. The bottom surface, the front wall, the rear wall, the side walls and/or the base body can thus be designed in one piece. In addition to decreased manufacturing costs, this can ensure an appealing appearance and/or a decreased cleaning effort.

(35) Sometimes, at least one bearing pedestal is provided in the base body for the connection to the second connecting element.

(36) The connection of the connecting module to the base body can be designed in such a way that the connecting module can be removed easily and without tools, and can thus be replaced. In addition to the removability and the associated handling and maintenance friendliness, it is also possible to create the option of exchanging or upgrading the component for influencing a pivoting movement of the lid with another component at a later point in time, without having to redesign the base body and/or the lid.

(37) In this way, the modularity can also provide different versions of the connecting module (for example, with motor element/without motor element and/or with brake/without brake), which are interchangeable among one another. In this respect, it is also possible to provide a system by way of the invention which comprises the base body, the lid and at least two connecting modules, wherein the connecting modules preferably each comprise a different component for influencing the pivoting movement of the lid.

(38) Individual features of the container according to the first aspect can be combined with features of the container according to the second aspect, and vice versa. In a preferred embodiment, the container according to the first aspect and the container according to the second aspect have the same features.

(39) In further specific embodiments, the container according to one of the above-described aspects can be part of a system including additional components. Additionally, a water bath vessel can be provided, which is accommodated by the base body. A space that is formed between the base body and the water bath vessel can be filled with water. A heating device may be provided to heat the water and/or keep it warm. The container can, for example, be arranged on a rack including a heating device therebeneath, wherein, for example, fuel cans or electrical heating elements can be used as the heating device.

Description

BRIEF DESCRIPTION OF THE DRAWING

(1) The invention will be described in more detail based on the accompanying figures. In the drawings:

(2) FIGS. 1-5 schematically show differing views of a container for receiving, keeping warm and/or serving food;

(3) FIG. 6 shows a detailed view of the container according to FIGS. 1-5 comprising a connecting module for connecting a lid to a base body;

(4) FIG. 7-9 schematically show differing views of the connecting module of FIG. 6;

(5) FIG. 10 shows a sectional view of the connecting module along an intersecting line A-A of FIG. 7 including a brake;

(6) FIG. 11 shows an exploded illustration of the brake of FIG. 6;

(7) FIGS. 12-13 show two further views of the connecting module of FIGS. 6-11;

(8) FIG. 14 shows a sectional view of a detail of the connecting module of FIG. 13 along an intersecting line B-B;

(9) FIG. 15 shows a sectional view of a detail of the connecting module of FIG. 12 along an intersecting line C-C;

(10) FIG. 16 shows a sectional view of a further brake including plotted flows of force;

- (11) FIG. 17 shows a sectional view of a further brake;
(12) FIG. 18 shows an exploded illustration of the connecting module of FIG. 17;
(13) FIG. 19 shows a sectional view of a further brake; and
(14) FIG. 20 shows different views of a form fit element attached to the lid.

DETAILED DESCRIPTION

(15) In the following description and in the figures, recurring and functionally equivalent features are denoted by the same reference numerals. For the sake of clarity, some of the reference numerals are not provided in every example, even though the associated elements may be present in the relevant example.

(16) FIGS. 1-5 show differing views of a container 1 for receiving, keeping warm and/or serving food. The container 1 comprises a trough-shaped (tub-shaped) base body 2 and a lid 10 for closing the base body 2, which can be pivoted about a pivot axis 40 between a closed position and an open position, the container 1 being shown in the closed state. The container 1 can optionally be arranged on a rack including a heating device therebeneath, wherein fuel cans or electrical heating elements are often used as the heating device. The base body 2 in general forms a receptacle for a likewise trough-shaped water bath vessel (not shown) in which food can be kept warm, wherein water is typically present between the base body 2 and the water bath vessel when the container 1 is in use so that the water bath vessel is situated in a water bath. The water is then heated by the aforementioned heating device. Such containers 1 are also known as “chafing dishes” and are used at buffets, for example, to keep food warm and showcase it. The container 1 is known in various embodiments, for example including flat or curved lids 10, which can be opened or closed as needed. To view the food even when the lid 10 is in the closed position, the lid can encompass a transparent region, which is formed by a glass panel or a plastic panel, for example.

(17) The base body 2 typically comprises a bottom surface 3, a front wall 4, a rear wall 5 and two side walls 6, 7 and is often designed in one piece. The rear wall 5 in general comprises a substantially flat or straight section, which is connected to the side walls 6, 7 via curved transition regions 8. The base body 2 is generally made of a metallic material, which is preferably acid-proof, rust-proof, tasteless, easy to clean and dishwasher-proof, wherein stainless steel, such as Cromargan®, represents a preferred material in many instances.

(18) Apart from the optional transparent region, the lid 10 can likewise be at least predominantly made of a metallic material, such as stainless steel, preferably Cromargan®. The lid 10 can comprise a handle 11, which facilitates the opening and closing of the lid 10 by hand.

(19) The following description addresses various aspects of the container 1. According to a first aspect, at least one brake 50 for braking (decelerating) a pivoting movement of the lid 10 and/or for holding the lid 10 in a position between a closed position and an open position is provided. According to a further aspect of the present invention, at least one connecting module 20 is provided, which connects the lid 10 to the base body 2. As described below, the brake 50 and the connecting module 20 can be combined with one another.

(20) First, the connecting module 20 and the features thereof will be addressed. An exemplary connecting module 20 is illustrated in greater detail in FIG. 6, which shows a detail of the container 1 of FIGS. 1-5. Additional views, sectional illustrations and exploded drawings of the connecting element 20 are shown in FIGS. 7-19.

(21) The connecting module 20 makes modular coupling of the lid 10 to the base body 2 possible. In particular, the connecting module 20 can be demounted both from the base body 2 and from the lid 10. Due to the modularity of the coupling, the elements 2, 10, 20 of the container 1 can be individually replaced, repaired and/or cleaned.

(22) The connecting module 20 comprises at least one component 50 for influencing a pivoting movement of the lid 10. The aforementioned component can in particular comprise or be the brake 50. As an alternative or in addition, the aforementioned component can comprise or be a motor element, which is not shown, for moving the lid 10 between the closed position and the open

position. The connecting module **20** can comprise a housing **17** comprising a first housing part **18** and a second housing part **19**, which can accommodate the aforementioned component **50**.

(23) The connecting module **20** furthermore comprises a first connecting element **21**, which is detachably connected to the lid **10**. The first connecting element **21** usually comprises a first form fit element **22**, which is designed as an elongated protrusion and can be connected to a first counter-piece **12**, which is attached to or integrally formed on the lid **10**, for forming a first form fit, wherein the counter-piece **12** is shaped in a manner that is complementary to the first form fit element **22**.

(24) Various views of the form fit element **12** or of the counter-piece **12** which is attached to or integrally formed on the lid **10** are shown in FIG. **20**. In the shown exemplary embodiment, the first counter-piece **12** includes a first elongated receptacle **13**, which accommodates the first form fit element **12** in a form-locked manner. The first connecting element **21** is non-rotatably connected to the lid **10** as a result of the first form fit and the elongated shape of the first form fit element **22**. A first locking depression **14** can be formed in the first receptacle **13** of the counter-piece.

(25) The first form fit element **22** can be integrally designed with the first housing part **18**. As an alternative, the first form fit element **22** can also be present as a separate component that is connected to the first housing part **18**. A longitudinal axis **41** of the first form fit element **22** can be oriented at an angle, for example perpendicularly, with respect to the pivot axis **40** of the lid **10**. An orientation of the longitudinal axis **41** in general furthermore depends on the instantaneous opening angle of the lid **10**.

(26) In addition, a first spring-loaded pressure piece **23** (spring plunger **23**) is provided, which latches in the first locking depression **14** when the first form fit is present. The first spring-loaded pressure piece **23** thus supports the retention of the first form fit element **22** in the first receptacle **13** of the counter-piece **12**. The first counter-piece **23** is arranged in a depression **28** that is formed in the first form fit element **22** and can be configured as a borehole. FIG. **7** contains a partial sectional illustration of the first pressure piece **23**. The pressure piece **23** in general comprises a sleeve **24**, a compression spring **25** and an element **26** preloaded by the compression spring **25**, which can be designed as a ball or pin, for example. The preloaded element **26** protrudes partially from the sleeve **24** and is usually held in the sleeve **24** by means of a protrusion **27** that is formed by the sleeve **24** and protrudes radially to the inside. The sleeve **24** is attached in the depression **28**. Optionally, the sleeve **24** may be omitted. In this case, the spring **25** and the preloaded element **26** are arranged directly in the depression **28**.

(27) A longitudinal axis of the first form fit element **22** and a longitudinal axis of the first receptacle **13** are often perpendicularly oriented with respect to the force of gravity when the lid **10** is closed. When the lid **10** has an opening angle of 90° , the aforementioned longitudinal axes of the elements **22**, **13** extend parallel to force of gravity. If the lid **10** is lifted in this position, the pressure piece **23** is released from the locking depression **14** so that the first form fit can be released, and the lid **10** can be removed from the connecting module **20**.

(28) In the shown example, the first form fit element **22** is thus accommodated by the first receptacle **13** of the counter-piece **12**. As an alternative, it may also be provided that the first form fit element accommodates the counter-piece (not shown). In this case, the first form fit element includes a first receptacle, which preferably has a locking depression. In this case, the counter-piece is furthermore configured as a protrusion, which can be accommodated by the first receptacle in a form-locked manner. The counter-piece attached to the lid can then comprise the first spring-loaded pressure piece.

(29) Furthermore, the connecting module **20** comprises a second connecting element **31**, which is detachably connected to the base body **2**.

(30) The second connecting element **31** is connected in a form-locked manner to a second counter-piece **9** provided at the base body **2**. For this purpose, the second connecting element **31** can comprise a second form fit element **32**, which can be connected to the second counter-piece **9** for

forming a second form fit. The second counter-piece **9** is shaped in a manner that is complementary to the second form fit element **32** and accommodates the second form fit element **32** in a form-locked manner. In the shown exemplary embodiment, the second counter-piece is implemented by a bearing pedestal **9**, wherein the second form fit element **32** is configured as a stud. Hereafter, reference is therefore made to the bearing pedestal **9** and the stud **32**. However, other complementary shapes are also conceivable for the second form fit. A longitudinal axis **42** of the stud, that is, of the second form fit element **32**, in general extends parallel to the pivot axis **40** of the lid **10**, preferably perpendicularly to the longitudinal axis **41**, and often times perpendicularly to the force of gravity.

(31) The bearing pedestal **9** usually has a second receptacle **15**, which accommodates the stud **32** in a form-locked manner. A second locking depression **16** can be formed in the inner wall of the second receptacle **15**.

(32) Furthermore, a second spring-loaded pressure piece **33** (spring plunger **33**) can be installed in the stud **32**, which latches in the second locking depression **16** when the second form fit is present. A longitudinal axis of the second spring-loaded pressure piece **33** usually extends at an angle with respect to the longitudinal axis **42** of the stud **32**, wherein a preferred angle can range between 80° and 100°, and in particular is approximately 90°.

(33) The second locking depression **16** can be configured as a through-opening in the bearing pedestal **9**, through which the second spring-loaded pressure piece **33** extends. As a result, a user has access from the outside to the second spring-loaded pressure piece **33** and can actuate this by pushing it with a finger. Thereafter, the second form fit can be released.

(34) Optionally, a third spring-loaded pressure piece **34** can be provided, which can be configured, on the one hand, to fix/center the second form fit when the second pressure piece **33** latches in the second locking depression **16** and, on the other hand, to support the release of the second form fit when the second pressure piece **33** is pushed out of the second locking depression **16**. The third spring-loaded pressure piece **34** is preferably likewise installed in the stud **32**, but can alternatively also be arranged in the second receptacle **15** in the bearing pedestal **9**. The third spring-loaded pressure piece **34** is preloaded in the latched state of the second spring-loaded pressure piece **33** and rests against an inner wall of the second receptacle **15**. Optionally, the third spring-loaded pressure piece **34** can engage in a third locking depression **35**, which is formed in the receptacle **15**. In one embodiment, which is likewise apparent in the figures, a longitudinal axis of the third spring-loaded pressure piece **34** extends parallel to the longitudinal axis **42** of the stud **32**, and thus perpendicularly to the longitudinal axis of the second spring-loaded pressure piece **33**. When the second pressure piece **33** is released from the second locking opening **16**, the third spring-loaded pressure piece **34** pushes the stud **32** at least partially out of the second receptacle **15**. In this way, the connecting module **20**, during the actuation of the second spring-loaded pressure piece **33**, moves out of the bearing pedestal **9** due to the spring force of the third spring-loaded pressure piece **34**, so as to provide haptic and/or acoustic feedback to the user that the connecting module **20** or the brake **50** has been released and is ready for removal.

(35) The second pressure piece **33** and/or the third pressure piece **34** can have a similar design as the first pressure piece **23**. For example, the respective pressure piece **33**, **34** can comprise a sleeve, a compression spring and an element preloaded by the compression spring, which can be designed as a ball or pin, for example. The preloaded element usually protrudes partially from the sleeve and is held in the sleeve by means of a protrusion that is formed by the sleeve and protrudes radially to the inside.

(36) In the shown example, the second form fit element **32** is thus accommodated by the second receptacle **15** of the bearing pedestal **9** in a form-locked manner. As an alternative, it may also be provided that the second form fit element accommodates the counter-piece (not shown). In this case, the second form fit element includes a second receptacle, which preferably has a second locking depression. In this case, the counter-piece is furthermore configured as a protrusion, which

can be accommodated by the second receptacle in a form-locked manner. The protrusion attached to or formed on the base body **2** can then comprise the second spring-loaded pressure piece and the third spring-loaded pressure piece.

(37) The connecting module **20** can comprise a rigid center piece **29**, which is connected to the first connecting element **21** and/or to the housing **17** at a first end and to the second connecting element **31** at an opposite end.

(38) Two connecting modules **20** are provided in the container **1** of FIGS. 1-5, which are arranged mirror-symmetrically with respect to one another. A plane of mirror symmetry typically extends parallel to the side walls **6**, **7**. The longitudinal axes **42** of the second form fit elements **32** configured as studs preferably point away from one another and are thus oriented anti-parallel to one another. So as to release the respective second form fit, the studs **32** thus must be pulled out of the second receptacles **15** in opposite directions. The longitudinal axes **41** of the first form fit elements **22** are preferably oriented parallel to one another and point in the same direction. By pulling on the lid or lifting the lid **10**, preferably at an opening angle of approximately 90°, the first form fits can be released simultaneously. As a result of the orientation of the longitudinal axes **41**, **42**, additionally the second form fit can only be released after the lid **10** has been demounted from the respective connecting module **20**. This prevents the connecting modules **20** from being inadvertently loaded into the dishwasher together with the lid **10**. In this way, the component **50** that is typically sensitive to water and/or cleaning agents can be protected and the service life thereof can be increased.

(39) As was already indicated above, each connecting module **20** is connected to the base body **2** by way of a bearing pedestal **9** that is attached, preferably welded, to the base body **2**. Each bearing pedestal **9** is preferably arranged at an upper side of the flat section of the rear wall **5**. In addition, the bearing pedestal **9** preferably abuts the transition region **8** and is thus positioned on the outside (on the right outside or left outside) of the rear wall **5**. The reason for the outwardly installed connection of the connecting module **20** is that the flexural rigidity of the side walls **6**, **7** and/or of the transition regions **8** is utilized. In this way, bending of the rear wall during opening and closing can be prevented, and additional rear wall reinforcements can be dispensed with (omitted).

(40) Hereafter, the brake **50** will be addressed in greater detail. The brake **50** configured as a friction brake is provided to decelerate the pivoting movement of the lid **10**, and to hold the lid **10** in a position between the closed position and the open position. By way of the brake **50**, the lid **10** can be held in a plurality of positions, preferably in an infinitely adjustable manner, between the closed position and the open position.

(41) The brake **50** comprises a freewheel **52**, which defines a blocking direction **44** and a freewheeling direction **45**. The blocking direction **44** corresponds to the closing direction of the lid **10** and/or the freewheeling direction **45** corresponds to the opening direction of the lid **10**. In other words, the freewheel **52** is coupled to a pivoting movement of the lid **10** in such a way that the brake **50** decelerates or holds the lid **10** in the closing direction, not, however, in the opening direction. The freewheel **52** can be configured as one-way roller clutch, for example.

(42) The lid **10** is coupled to a rotatable shaft **54**, which defines an axis of rotation or pivot axis **40**. The shaft **54** can have a non-circular protrusion **56** that extends in the axial direction and engages in a corresponding recess **57** in the first housing part **18** in a form-locked manner. An attachment means, such as a screw **55**, can be provided to attach the shaft **54** to the first housing part **18**.

(43) Optionally, a bearing bushing **58** can be provided, which surrounds the shaft **54**. The bearing bushing **58** can define an interior space of the brake **50**, in which the freewheel **52** is situated. The freewheel **52** is preferably arranged radially between the bearing bushing **58** and the shaft **54**. In the shown example, the freewheel **52** is arranged directly on the shaft **54**, and thus no further components are present between the freewheel **52** and the shaft **54**. Furthermore, two rolling bearings **60** are provided, which are arranged on both sides of the freewheel **52** on the shaft **54**. The rolling bearings **60** can be designed as grooved ball bearings, for example, and are generally

arranged radially between the bearing bushing **58** and the shaft **54**. The rolling bearings **60** enable a rotation of the shaft **54** relative to the bearing bushing **58**.

(44) Optionally, the bearing bushing **58** can have a circumferential, radially inwardly directed protrusion **61**, which in the axial direction often abuts the rolling bearings **60** on the one hand, and the freewheel **52** on the other hand. Two retaining rings **66** can be provided, which each abut the ball bearings **60** and form an axial delimitation for the ball bearings **60**. The bearing bushing **58** can furthermore have a shoulder **59**, which points radially outwardly. The rolling bearings **60** are likewise arranged in the interior space defined by the bearing bushing **58**.

(45) The shaft **54** can have a plurality of steps, which extend in the radial direction and are arranged next to one another in the axial direction. The elements **53**, **50** and **66** can each be arranged on a step having differing radial heights.

(46) A first cover **62** can be provided for covering the interior space of the brake **50** surrounded by the bearing bushing **58**. The bearing bushing **58** can be non-rotatably connected to the first cover **62**, for example by means of screws **63**, which engage in the shoulder **59**, for example. For sealing, a sealing element **64** such as an O-ring, which can be formed of an elastic material, can be provided between the bearing bushing **58** and the first cover **62**. For further sealing, a further sealing element, such as a radial shaft seal ring **65**, can be provided radially between the cover **62** and the shaft **54**. The radial shaft seal ring **65** defines a region **53** which is filled with lubricant, such as grease, for lubrication and further sealing (refer to FIG. **16**).

(47) Due to the O-ring **64** and the radial shaft seal ring **65**, the interior space of the brake **50**, in which the one-way roller clutch **52** and the ball bearings **60** are situated, is sufficiently protected against water and moisture.

(48) The bearing bushing **58** can be connected to a second cover **68** on the axially other side. The first cover **62** and the second cover **68** can thus be arranged on axially opposing sides of the bearing bushing **58**. The second cover **68** preferably has an internal thread for this purpose, which can be screwed to an external thread of the bearing bushing **58**. A spacer ring **67** or a spacer sleeve can be arranged axially between the second cover **68** and the bearing bushing. A retaining ring O-ring **69** can be provided axially between the second cover **68** and the second housing part **19**. The O-ring is used for sealing against liquids from the outside, which other-wise could penetrate into the element **68** through the small thread. Preferably, for example, a sealing paste can be provided at least in regions between the elements **68**, **67** and **58** so as to seal this region against moisture from the outside.

(49) The first cover **62** and the second cover **68**, together with the bearing bushing **58**, can be interpreted as an inner housing, while the first housing part **18** and the second housing part **19** can be interpreted as an outer housing.

(50) FIG. **11** shows an exploded illustration of the brake **50** or of the connecting module **20**. It is apparent in FIG. **11** that the bearing bushing **58** radially on the outside, in the circumferential direction, typically comprises a round section and a flat section. The flat section of the bearing bushing **58** is in particular suitable for a rotationally fixed form fit with components arranged on the bearing bushing **58**.

(51) The brake **50** can moreover comprise at least one brake pad **70**, preferably two brake pads **70**, and a brake disk **72**. The term “brake disk” used in this specification was selected analogously to a disk brake of a vehicle and due to the disk shape of the element **72**. Deviating from a disk brake used in a vehicle, however, the brake pads **70** rotate, while the brake disk **72** is at static. Instead of the term “brake disk,” it is also possible to use wording such as “brake lever” or “brake arm.”

(52) The brake pads **70** are usually arranged on the bearing bushing **58** and can comprise a flat section on the insides thereof so as to enable a rotationally fixed form fit with the bearing bushing **58**. The brake disk **72** is in general fixedly connected to the rigid center piece **29**, and is therefore arranged so as to be static with respect to the base body **2**. The brake pads **70** are arranged in the axial direction on both sides of the brake disk **72**. A spring element **74** is preferably provided,

which pushes the brake pads **70** and the brake disk **72** together in the axial direction. The spring element **74** can be configured as a wave spring, for example. An adjusting washer **73**, which distributes a spring force of the spring element **74** evenly on the brake pad **70**, can be arranged between the brake pad **70** and the spring element **74**. A further washer **75**, which rests against an end face of the second cover **68**, can be arranged with form fit on the axially other side of the spring element **74**. The purpose of this washer **75** is to prevent the wave spring **74** from becoming twisted when the second cover **68** is screwed on. A sliding bearing **76** can be arranged radially between the brake disk **72** and the bearing bushing **28**, which allows the bearing bushing **58** to be rotated relative to the brake disk **72**.

(53) The largest possible opening angle of the lid **10** can be established by a stop **48** present at the center piece **29**, wherein the stop **48** is sometimes referred to as a nose. This stop **48** thus limits the pivoting movement of the lid **10**. The stop **48** engages in a slot-shaped recess **49**, which runs, for example, axially between the brake disk **72** and the first housing part **18** and extends in the circumferential direction.

(54) The mechanism of action of a similar brake **50** is described hereafter based on FIG. **16**. FIG. **16** shows a section of a brake **50** and of a connecting module **20**, which are attached at the left of the container **1**, while the brake **50** and the connecting module **20** of FIG. **10** are attached at the right of the container **1**. Moreover, the brake **50** of FIG. **16** is designed axially slightly shorter than the brake **50** of FIG. **10**. Functionally equivalent features are denoted by identical reference numerals, and a repetition of the description is dispensed with here.

(55) Two flows of force are plotted in FIG. **16** by means of arrows **46**, **47**. The arrows **47** indicate the flow of force when the lid **10** is being opened, while the arrows **46** indicate the flow of force when the lid **10** is being closed.

(56) As described above, the lid **10** is non-rotatably connected to the first housing part **18** and to the shaft **54** by way of the first form fit element **22**. The opening direction of the lid **10** corresponds to the freewheeling direction **45** of the shaft **54** and of the freewheel **52**, so that the shaft **54** rotates along when the lid **10** is being opened. When the lid **10** is being opened, the torque is thus not passed on to the braking mechanism, and the brake **50** is thus not actuated in this case. In this way, in particular the components **19**, **58**, **68**, **70** do not rotate along. The at least one brake pad **70** and the brake disk **72** are static relative to one another in the freewheeling direction **45** of the shaft **54**.

(57) If, conversely, the lid **10** is rotated from an open position into a closed position, the rotation of the shaft **54** is transferred from the freewheel **52** by way of force fit to the bearing bushing **58**. The torque is conducted from the bearing bushing **58** into the brake pads **70** by means for form fit. The brake pads **70** are pushed by the spring element **74** against the axial end faces of the brake disk **72**. In the blocking direction **44** of the shaft **54**, a relative movement of the brake pad **70** and of the brake disk **72** is decelerated (braked), and the braking action is thus generated by the friction between the brake pads **70** and the brake disk **72**. Overall, the rotation of the shaft **54** and the pivoting movement of the lid **10** are thus decelerated. The force exerted on the brake disk **72** flows off via the center piece **29** and the second connecting element **21** onto the bearing pedestal **9** and the container **1**. If no manual force is exerted, and only the force of gravity acts on the lid **10**, the brake **50** ensures that the lid **10** is held in the position thereof. The braking force is preferably designed in such a way that the lid **10** is held in an infinitely variable manner and without play in each position between the closed position and the open position. This also means that a minor manual force for overcoming the holding force of the brake **50** is required for closing the lid **10**. This required manual force is the greatest in a completely open position at approximately 90°, and steadily decreases when the lid **10** is being closed due to the growing influence of the force of gravity.

(58) The sliding bearing **76** arranged between the brake disk **72** and the bearing bushing **58** allows a relative rotational movement of the bearing bushing **58** relative to the brake disk **72**. When the lid **10** is being closed, amongst others the components **19**, **58**, **68**, **70** and the components non-

rotatably connected thereof thus also rotate along.

(59) FIG. 17 shows a longitudinal sectional view of a similar brake 50, which has a simpler design than the brake 50 shown in the sectional illustration in FIG. 10. In FIG. 17, identical features are denoted by identical reference numerals, and a repetition of the description is dispensed with here. (60) The brake 50 comprises a freewheel 51, which is configured as a one-way roller clutch and in which two rolling bearings are integrated. In this way, the separate rolling bearings 60 of FIG. 10 can be dispensed with. Moreover, the first cover 62 is dispensed with in the embodiment of FIG. 17. An optional radial shoulder 30 of the shaft 54 delimits the freewheel 51 in the axial direction, so that a retaining ring 66 can also be dispensed with in this region. On the other side of the shaft 54, an adjusting washer 77 is arranged axially between the retaining ring 66 and the freewheel 51. Overall, the brake 50 of FIG. 17 thus requires fewer components than the brake 50 of FIG. 10, see also the exploded drawing of the brake 50 in FIG. 18.

(61) Another embodiment of the brake 50 is shown in FIG. 19, which discloses a longitudinal sectional view of the brake 50.

(62) The brake of FIG. 19 resembles the brake 50 of FIG. 17 to the effect that the rolling bearings are integrated into the freewheel 51, and the first cover 62 of the brake 50 of FIG. 10 was dispensed with. Instead of the radial shaft seal ring 65 of FIG. 17, however, an optional sealing ring 78 is provided in the brake of FIG. 19, which seals the system radially to the outside. The sealing ring 78 is typically located between the brake disk 72 and the first partial housing 18. The sealing ring 78 is preferably recessed in a groove 79 of the brake disk 72. A further sealing ring 78 can be arranged between the brake disk 72 and the second partial housing 19, preferably likewise in a groove 79 of the brake disk 72. Since the lateral surfaces of the partial housings 18, 19 which make contact with one another move relative radially with respect to the seals 78, quad rings can optionally be selected for the seals 78, which in general have a lubricant reservoir, whereby friction is minimized. In the region of the screw 55 and/or between the shaft 56 and the bearing shoulder of the first housing part 18, a sealing paste and/or an additional sealing element can also be provided for sealing so as to shield the system against moisture from the outside.

(63) Another special feature of the embodiment of FIG. 19 is that the second cover 68 was dispensed with. Instead, the second housing part 19 is directly connected to the bearing bushing 58, typically by way of a screw connection. In this way, fewer parts are needed.

LIST OF REFERENCE NUMERALS

(64) 1 container 2 base body 3 bottom surface 4 front wall 5 rear wall 6 side wall 7 side wall 8 transition region 9 bearing pedestal 10 lid 11 handle 12 first counter-piece 13 first receptacle 14 first locking depression 15 second receptacle 16 second locking depression 17 housing 18 first housing part 19 second housing part 20 connecting module 21 first connecting element 22 first form fit element 23 first spring-loaded pressure piece (spring plunger) 24 sleeve 25 compression spring 26 preloaded element 27 protrusion 28 depression 29 center piece 30 radial shoulder 31 second connecting element 32 second form fit element 33 second spring-loaded pressure piece (spring plunger) 34 third spring-loaded pressure piece (spring plunger) 35 third locking depression 40 pivot axis of the lid 41 longitudinal axis of the first form fit element 22 42 longitudinal axis of the second form fit element 32 44 blocking direction 45 freewheeling direction 46 flow of force when closing of the lid 47 flow of force when opening of the lid 48 stop 49 slot-shaped recess 50 brake 51 freewheel 52 freewheel 53 region filled with lubricant 54 shaft 55 attachment means 56 axial protrusion 57 recess 58 bearing bushing 59 shoulder 60 rolling bearing 61 protrusion 62 first cover 63 screw 64 O-ring 65 radial shaft seal ring 66 retaining ring 67 spacer ring 68 second cover 69 O-ring 70 brake pad 72 brake disk 73 adjusting washer 74 spring element 75 disk with form fit 76 sliding bearing 77 adjusting washer 78 quad ring 79 groove

Claims

1. A container, in particular for receiving, keeping warm and/or serving food, comprising: a trough-shaped base body; a lid for closing the base body configured to be pivoted between a closed position and an open position; and a connecting module for connecting the lid to the base body, the connecting module comprising a first connecting element, which is detachably connected to the lid, the connecting module comprising a second connecting element, which is detachably connected to the base body, the connecting module comprising at least one component for influencing a pivoting movement of the lid, wherein the first connecting element comprises a first form fit element, configured to be connected to a first counter-piece attached to the lid for forming a first form fit, wherein a first spring plunger is provided, which latches in a first locking depression when the first form fit is present.
2. The container according to claim 1, wherein the first spring plunger is installed in the first form fit element, and the first locking depression is formed in the first counter-piece.
3. The container according to claim 1, wherein, at a certain opening angle of the lid, a longitudinal axis of the first form fit element and a longitudinal axis of the first counter-piece extend parallel to a force of gravity, so that the form-locked connection between the lid and the first connecting element is configured to be released by lifting of the lid.
4. The container according to claim 1, wherein the second connecting element comprises a second form fit element, configured to be connected to a second counter-piece, which is attached to the lid, for forming a second form fit, wherein a second spring plunger is provided, which latches in a second locking depression when the second form fit is present.
5. The container according to claim 4, wherein the second form fit element engages in a second receptacle formed in the second counter-piece, the second locking depression being formed in an inner wall of the second receptacle, the second spring plunger being installed in the second form fit element and extending at an angle, with respect to a longitudinal axis of the second form fit element.
6. The container according to claim 4, wherein a third spring plunger, which is preloaded when the second spring plunger is latched and which is configured to push the second form fit element at least partially out of the second receptacle when a second pressure piece is released from the second locking depression, is provided for releasing the second form fit, and wherein the third spring plunger is installed in the second form fit element and rests against the inner wall of the second receptacle when the second spring plunger is latched.
7. The container according to claim 1, wherein the connecting module comprises a housing for accommodating the at least one component.
8. The container according to claim 1, wherein at least one bearing pedestal is provided at the base body for connection to the second connecting element.
9. The container according to claim 1, wherein the base body comprises a bottom surface, a front wall, a rear wall and two side walls.
10. The container according to claim 1, the at least one component for influencing the pivoting movement of the lid comprising: a brake for braking the pivoting movement of the lid and/or for holding the lid in a position between the closed position and the open position and/or a motor element for moving the lid between the closed position and the open position.
11. A container, in particular for receiving, keeping warm and/or serving food, comprising: a trough-shaped base body; a lid for closing the base body configured to be pivoted between a closed position and an open position; and at least two connecting modules for connecting the lid to the base body, each connecting module comprising: a first connecting element, which is detachably connected to the lid, a second connecting element, which is detachably connected to the base body, and at least one component for influencing a pivoting movement of the lid, wherein the first connecting element comprises a first form fit element, configured to be connected to a first counter-piece attached to the lid for forming a first form fit, wherein a first spring plunger is provided,

which latches in a first locking depression when the first form fit is present, and wherein the second connecting element comprises a second form fit element, configured to be connected to a second counter-piece, which is attached to the lid, for forming a second form fit, wherein a second spring plunger is provided, which latches in a second locking depression when the second form fit is present, the longitudinal directions of the first form fit elements extending parallel to one another and pointing in the same direction, and the longitudinal directions of the second form fit elements extending parallel to one another, but being oriented in opposite directions.

12. The container according to claim 9, wherein the rear wall extends in a lateral direction from a first end section to a second end section, the end sections being each connected to one of the two side walls, and the connecting modules being attached to the rear wall in the end sections.

13. The container according to claim 9, wherein the connecting module is attached to an upper side of the rear wall of the base body.

14. A container, in particular for receiving, keeping warm and/or serving food, comprising: a trough-shaped base body; a lid for closing the base body configured to be pivoted between a closed position and an open position; and a connecting module for connecting the lid to the base body, the connecting module comprising a first connecting element, which is detachably connected to the lid, the connecting module comprising a second connecting element, which is detachably connected to the base body, the connecting module comprising at least one component for influencing a pivoting movement of the lid, wherein the second connecting element comprises a second form fit element, configured to be connected to a second counter-piece, which is attached to the lid, for forming a second form fit, wherein a second spring plunger is provided, which latches in a second locking depression when the second form fit is present.

15. The container according to claim 14, wherein the second form fit element engages in a second receptacle formed in the second counter-piece, the second locking depression being formed in an inner wall of the second receptacle, the second spring plunger being installed in the second form fit element and extending at an angle with respect to a longitudinal axis of the second form fit element.

16. The container according to claim 14, wherein a third spring plunger, which is preloaded when the second spring plunger is latched and which is configured to push the second form fit element at least partially out of the second receptacle when a second pressure piece is released from the second locking depression, is provided for releasing the second form fit, and wherein the third spring plunger is installed in the second form fit element and rests against the inner wall of the second receptacle when the second spring plunger is latched.

17. The container according to claim 14, wherein the connecting module comprises a housing for accommodating the at least one component.

18. The container according to claim 14, wherein at least one bearing pedestal is provided at the base body for connection to the second connecting element.

19. The container according to claim 14, wherein the base body comprises a bottom surface, a front wall, a rear wall and two side walls.

20. The container according to claim 14, the at least one component for influencing the pivoting movement of the lid comprising: a brake for braking the pivoting movement of the lid and/or for holding the lid in a position between the closed position and the open position and/or a motor element for moving the lid between the closed position and the open position.
